Chapter 11. Bayesian Inference

Prior and posterior, Bayesian inference, Bayes and minimax estimators

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11.1 Prior and posterior distributions

The Bayes criteria of Chapters 8 and 9 weight the parameter values by a prior. Bayesian inference takes this literally: before the data are seen, uncertainty about $\theta \in \Omega$ is described by a **prior density** $\pi(\theta)$. Whether such a prior exists, and whether it is unique, is debatable, and this is the source of the long-standing controversy about Bayesian methods. Here we take the prior as given. To stress conditioning we write $f(x \mid \theta)$ for the model density.

Theorem 11.1 (Bayes' theorem)

If $X_1, \dots, X_n$ is a sample from $f(x \mid \theta)$ and θ has prior density $\pi(\theta)$, then given $\mathbf{X} = \mathbf{x}$ the density of θ is

$$\pi(\theta \mid \mathbf{x}) \propto L(\theta; \mathbf{x})\pi(\theta), \qquad L(\theta; \mathbf{x}) = \prod_{i=1}^{n} f(x_i \mid \theta).$$

Proof. By the definition of conditional density, $\pi(\theta \mid \mathbf{x}) = f(\mathbf{x}, \theta)/f_{\mathbf{X}}(\mathbf{x}) = f(\mathbf{x} \mid \theta)\pi(\theta)/f_{\mathbf{X}}(\mathbf{x})$. The normalizing constant is the marginal density $f_{\mathbf{X}}(\mathbf{x}) = \int_\Omega f(\mathbf{x} \mid \theta)\pi(\theta)\, d\theta$ (a sum in the discrete case), which does not involve θ. ■

$\pi(\theta \mid \mathbf{x})$ is the **posterior density**; in words, posterior $\propto$ likelihood $\times$ prior. A prior that is not a probability density ($\int \pi = \infty$) is called **improper**; it can still be used if the posterior is proper.

Example 11.1 (Binomial–beta)

Bernoulli(θ) data with prior Beta(α, β): with $t = \sum x_i$,

$$\pi(\theta \mid \mathbf{x}) \propto \theta^t (1-\theta)^{n-t} \theta^{\alpha-1} (1-\theta)^{\beta-1},$$

so $\theta \mid \mathbf{x} \sim \text{Beta}(t+\alpha, n-t+\beta)$.

Example 11.2 (Poisson–gamma)

Poisson(θ) data with prior Gamma(α, β): $\pi(\theta \mid \mathbf{x}) \propto e^{-n\theta}\theta^t \theta^{\alpha-1} e^{-\theta/\beta}$, so $\theta \mid \mathbf{x} \sim \text{Gamma}(t+\alpha, (n+\beta^{-1})^{-1})$.

Example 11.3 (Exponential–gamma)

Exp(θ) data, with the rate $\lambda = 1/\theta$ given a Gamma(α, β) prior: $\pi(\lambda \mid \mathbf{x}) \propto \lambda^n e^{-\lambda t} \lambda^{\alpha-1} e^{-\lambda/\beta}$, so $\lambda \mid \mathbf{x} \sim \text{Gamma}(n+\alpha, (t+\beta^{-1})^{-1})$, i.e. $(t+\beta^{-1})/\theta \mid \mathbf{x} \sim \text{Gamma}(n+\alpha, 1)$. The posterior density of θ itself follows by the change of variables $\lambda = 1/\theta$: $\pi(\theta \mid \mathbf{x}) = \pi(\lambda \mid \mathbf{x})\theta^{-2}$.

Example 11.4 (Normal mean, known variance)

$N(\theta, \sigma^2)$ data, σ^2 known, prior $N(\mu_0, \sigma_0^2)$:

$$\pi(\theta \mid \mathbf{x}) \propto \exp\Big\{ -\frac{n}{2\sigma^2}(\theta - \bar{x})^2 - \frac{1}{2\sigma_0^2}(\theta - \mu_0)^2 \Big\}.$$

Completing the square, $\theta \mid \mathbf{x} \sim N(\mu(\mathbf{x}), \sigma^2(\mathbf{x}))$ with

$$\mu(\mathbf{x}) = \frac{n\sigma^{-2}\bar{x} + \sigma_0^{-2}\mu_0}{n\sigma^{-2} + \sigma_0^{-2}}, \qquad \sigma^2(\mathbf{x}) = (n\sigma^{-2} + \sigma_0^{-2})^{-1}.$$

Posterior precision is data precision plus prior precision.

In these examples the posterior belongs to the same family as the prior; such a prior is a **conjugate prior**.

Theorem 11.2 (Normal model, conjugate prior)

Let $X_1, \dots, X_n$ be $N(\mu, \sigma^2)$, and let the prior of $\theta = (\mu, \sigma^2)$ be

$$\mu \mid \sigma^2 \sim N(\mu_0, \sigma^2/\kappa_0), \qquad \nu_0\sigma_0^2/\sigma^2 \sim \chi^2(\nu_0).$$

Put $\kappa_n = n + \kappa_0$, $\nu_n = n + \nu_0$, $s^2 = \sum(x_i - \bar{x})^2/(n-1)$, and

$$\mu_n(\mathbf{x}) = \frac{n\bar{x} + \kappa_0\mu_0}{n + \kappa_0}, \qquad \nu_n\sigma_n^2(\mathbf{x}) = \nu_0\sigma_0^2 + (n-1)s^2 + (\kappa_0^{-1} + n^{-1})^{-1}(\bar{x} - \mu_0)^2.$$

Then

(a) $\mu \mid \sigma^2, \mathbf{x} \sim N(\mu_n(\mathbf{x}), \sigma^2/\kappa_n)$ and $\nu_n\sigma_n^2(\mathbf{x})/\sigma^2 \mid \mathbf{x} \sim \chi^2(\nu_n)$;

(b) $\dfrac{\mu - \mu_n(\mathbf{x})}{\sigma_n(\mathbf{x})/\sqrt{\kappa_n}} \,\Big|\, \mathbf{x} \sim t(\nu_n)$.

Proof. (a) Work with μ and $\tau = \sigma^{-2}$. The likelihood and prior are

$$L \propto \tau^{n/2} \exp\Big\{ -\frac{\tau}{2}\big[(n-1)s^2 + n(\mu - \bar{x})^2\big]\Big\},$$

$$\pi(\mu \mid \tau) \propto \tau^{1/2} e^{-\frac{\tau\kappa_0}{2}(\mu-\mu_0)^2}, \qquad \pi(\tau) \propto \tau^{\nu_0/2-1} e^{-\frac{\tau}{2}\nu_0\sigma_0^2}.$$

Completing the square in μ,

$$n(\mu - \bar{x})^2 + \kappa_0(\mu - \mu_0)^2 = \kappa_n(\mu - \mu_n(\mathbf{x}))^2 + \frac{n\kappa_0}{n + \kappa_0}(\bar{x} - \mu_0)^2,$$

and $\frac{n\kappa_0}{n+\kappa_0} = (\kappa_0^{-1} + n^{-1})^{-1}$. Hence

$$\pi(\mu, \tau \mid \mathbf{x}) \propto \tau^{1/2} e^{-\frac{\tau\kappa_n}{2}(\mu-\mu_n(\mathbf{x}))^2} \cdot \tau^{\nu_n/2-1} e^{-\frac{\tau}{2}\nu_n\sigma_n^2(\mathbf{x})},$$

which is the stated normal–chi-square form.

(b) By (a), given $\mathbf{x}$, $Z = \frac{\mu - \mu_n(\mathbf{x})}{\sigma/\sqrt{\kappa_n}} \sim N(0,1)$ conditionally on σ^2; this law does not involve σ^2, so Z is independent of $V = \nu_n\sigma_n^2(\mathbf{x})/\sigma^2 \sim \chi^2(\nu_n)$. The ratio $Z/\sqrt{V/\nu_n}$ is the stated quantity, which is $t(\nu_n)$. ∎

Without conjugacy, the marginal $f_{\mathbf{X}}(\mathbf{x})$ is often intractable, and the posterior is computed by Markov chain Monte Carlo. Since the theory is cleanest with conjugate priors, we use them below.

11.2 Bayesian inference

Bayesian inference summarizes the posterior. Point estimates are the **posterior mean**, **median** or **mode**; precision is measured by the posterior variance or quantiles. An interval $I(\mathbf{x})$ with $P(\theta \in I(\mathbf{x}) \mid \mathbf{X} = \mathbf{x}) = 1 - \alpha$ is a $1-\alpha$ **credible interval**. Unlike a confidence interval, its probability statement is about θ given the observed data.

Example 11.5 (Normal mean, known variance)

In Example 11.4 the posterior is normal, so its mean, median and mode coincide:

$$\hat{\theta} = \frac{n\sigma^{-2}\bar{x} + \sigma_0^{-2}\mu_0}{n\sigma^{-2} + \sigma_0^{-2}},$$

a weighted average of the prior mean and $\bar{x}$, with weights equal to the prior precision σ_0^{-2} and the data precision $n\sigma^{-2}$. The $1-\alpha$ credible interval is $\hat{\theta} \pm z_{\alpha/2}\sigma(\mathbf{x})$. For instance, with $\sigma = 10$, $n = 25$, $\bar{x} = 104$ and prior $N(100, 5^2)$: $n\sigma^{-2} = 0.25$, $\sigma_0^{-2} = 0.04$, $\hat{\theta} = (26 + 4)/0.29 = 103.45$, $\sigma(\mathbf{x}) = 0.29^{-1/2} = 1.857$, and the 95% credible interval is 103.45 ± 3.64.

Example 11.6 (Normal model, conjugate prior)

By Theorem 11.2(b), when $\nu_n > 1$ the posterior of μ is a shifted, scaled t, so the posterior mean, median and mode equal $\hat{\mu} = \mu_n(\mathbf{x})$, and the $1 - \alpha$ credible interval is $\hat{\mu} \pm t_{\alpha/2}(\nu_n)\sigma_n(\mathbf{x})/\sqrt{\kappa_n}$.
For σ^2, $Y = \nu_n\sigma_n^2(\mathbf{x})/\sigma^2 \mid \mathbf{x} \sim \chi^2(\nu_n)$ gives

$$\hat{\sigma}^{2,\text{mean}} = \nu_n\sigma_n^2(\mathbf{x})E[Y^{-1}] = \frac{\nu_n}{\nu_n - 2}\sigma_n^2(\mathbf{x}), \qquad \hat{\sigma}^{2,\text{median}} = \frac{\nu_n}{\chi^2_{0.5}(\nu_n)}\sigma_n^2(\mathbf{x}),$$

and, from the posterior density of σ^2, $\hat{\sigma}^{2,\text{mode}} = \frac{\nu_n}{\nu_n+2}\sigma_n^2(\mathbf{x})$. As $n \to \infty$ all three factors tend to 1. The $1 - \alpha$ credible interval is

$$\frac{\nu_n\sigma_n^2(\mathbf{x})}{\chi^2_{\alpha/2}(\nu_n)} \le \sigma^2 \le \frac{\nu_n\sigma_n^2(\mathbf{x})}{\chi^2_{1-\alpha/2}(\nu_n)}.$$

Example 11.7 (Binomial–beta)

With a $\text{Beta}(\alpha_1, \alpha_2)$ prior, $\theta \mid \mathbf{x} \sim \text{Beta}(t + \alpha_1, n - t + \alpha_2)$. The posterior mean is

$$\hat{\theta}^{\text{mean}} = \frac{t + \alpha_1}{n + \alpha_1 + \alpha_2} = \frac{n}{n + \alpha_1 + \alpha_2} \cdot \frac{t}{n} + \frac{\alpha_1 + \alpha_2}{n + \alpha_1 + \alpha_2} \cdot \frac{\alpha_1}{\alpha_1 + \alpha_2},$$

again a weighted average of the sample proportion and the prior mean, and the mode is $\frac{(t+\alpha_1-1)_+}{(t+\alpha_1-1)_+ + (n-t+\alpha_2-1)_+}$. With $B_p(r_1, r_2)$ the upper p quantile of $\text{Beta}(r_1, r_2)$, the credible interval is $[B_{1-\alpha/2}, B_{\alpha/2}]$ for $\text{Beta}(t+\alpha_1, n-t+\alpha_2)$. Since $\text{Beta}(r_1, r_2) \overset{d}{=} \frac{(r_1/r_2)F}{1+(r_1/r_2)F}$ with $F \sim F(2r_1, 2r_2)$, these quantiles come from F tables. For large n, the posterior is approximately $N(\hat{\theta}, \hat{\theta}(1-\hat{\theta})/(n+\alpha_1+\alpha_2))$, giving the approximate interval $\hat{\theta} \pm z_{\alpha/2}\sqrt{\hat{\theta}(1-\hat{\theta})/(n + \alpha_1 + \alpha_2)}$. For example, $t = 12$ successes in $n = 40$ trials with a uniform prior ($\alpha_1 = \alpha_2 = 1$) give $\hat{\theta}^{\text{mean}} = 13/42 = 0.310$ and the approximate 95% interval 0.310 ± 0.140.

Bayesian testing. To decide between $H_0 : \theta \in \Omega_0$ and $H_1 : \theta \in \Omega_1$, compare the posterior probabilities through the **posterior odds**

$$\frac{P(\theta \in \Omega_0 \mid \mathbf{X} = \mathbf{x})}{P(\theta \in \Omega_1 \mid \mathbf{X} = \mathbf{x})} = \frac{P(\theta \in \Omega_0 \mid \mathbf{X} = \mathbf{x})}{1 - P(\theta \in \Omega_0 \mid \mathbf{X} = \mathbf{x})}.$$

Example 11.8 (Simple hypotheses)

For $H_0 : \theta = \theta_0$ vs $H_1 : \theta = \theta_1$ with prior probabilities π_0, π_1,

$$\frac{P(\theta = \theta_0 \mid \mathbf{x})}{P(\theta = \theta_1 \mid \mathbf{x})} = \frac{f(\mathbf{x} \mid \theta_0)\pi_0}{f(\mathbf{x} \mid \theta_1)\pi_1}.$$

Rejecting H_0 when the posterior odds are below 1 is exactly the Bayes test of Theorem 9.1. For $N(\mu, 1)$ data and μ_0 vs μ_1, the posterior odds are $\exp\{n(\mu_0 - \mu_1)(\bar{x} - \frac{\mu_0+\mu_1}{2}) + \log(\pi_0/\pi_1)\}$.

Example 11.9 (One-sided hypotheses)

(a) In the normal model with conjugate prior, for $H_0 : \mu \le \mu_0$ vs $H_1 : \mu > \mu_0$, Theorem 11.2(b) gives

$$P(\mu \le \mu_0 \mid \mathbf{x}) = F_{t(\nu_n)}\Big(\frac{\mu_0 - \mu_n(\mathbf{x})}{\sigma_n(\mathbf{x})/\sqrt{\kappa_n}}\Big),$$

with $F_{t(\nu)}$ the $t(\nu)$ cdf, and the posterior odds follow. (b) In the binomial–beta model, for $H_0 : \theta \le \theta_0$, $P(\theta \le \theta_0 \mid \mathbf{x})$ is the $\mathrm{Beta}(t + \alpha_1, n - t + \alpha_2)$ cdf at θ_0, computable from F tables, or for large n approximately $\Phi((\theta_0 - \hat\theta)/\sqrt{\hat\theta(1-\hat\theta)/(n + \alpha_1 + \alpha_2)})$.

11.3 Bayes and minimax estimators

Bayesian inference uses only the observed data, through the posterior. Frequentist inference instead evaluates a procedure over repeated samples, e.g. by its MSE $E[(\hat\theta(\mathbf{X}) - \theta)^2 \mid \theta]$. As in Chapter 8, two criteria free of the unknown θ are the Bayes MSE and the maximum MSE:

$$r(\pi, \hat\theta) = \int_\Omega E[(\hat\theta - \theta)^2 \mid \theta]\pi(\theta)\, d\theta, \qquad \max_{\theta \in \Omega} E[(\hat\theta - \theta)^2 \mid \theta].$$

Definition (Bayes and minimax estimators)

(a) $\hat\theta^\pi$ is a **Bayes estimator** for the prior π if $r(\pi, \hat\theta^\pi) = \min_{\hat\theta} r(\pi, \hat\theta)$. (b) $\hat\theta^m$ is a **minimax estimator** if $\max_\theta E[(\hat\theta^m - \theta)^2 \mid \theta] = \min_{\hat\theta} \max_\theta E[(\hat\theta - \theta)^2 \mid \theta]$.

Theorem 11.3 (Bayes estimator under squared error)

The Bayes estimator minimizes the posterior expected squared error and is the posterior mean:

$$\hat\theta^\pi(\mathbf{x}) = \arg\min_a E[(a - \theta)^2 \mid \mathbf{X} = \mathbf{x}] = E(\theta \mid \mathbf{X} = \mathbf{x}).$$

Proof. $r(\pi, \hat\theta)$ is the expectation of $(\hat\theta(\mathbf{X}) - \theta)^2$ under the joint law of $(\mathbf{X}, \theta)$, so by iterated expectation

$$r(\pi, \hat\theta) = E\big[E\{(\hat\theta(\mathbf{X}) - \theta)^2 \mid \mathbf{X}\}\big].$$

If $\hat\theta^\pi(\mathbf{x})$ minimizes the inner conditional expectation for each $\mathbf{x}$, then $E\{(\hat\theta(\mathbf{X}) - \theta)^2 \mid \mathbf{X}\} \ge E\{(\hat\theta^\pi(\mathbf{X}) - \theta)^2 \mid \mathbf{X}\}$ for every $\hat\theta$, and taking expectations gives $r(\pi, \hat\theta) \ge r(\pi, \hat\theta^\pi)$. Finally, $E[(a - \theta)^2 \mid \mathbf{x}] = \mathrm{Var}(\theta \mid \mathbf{x}) + (a - E[\theta \mid \mathbf{x}])^2$ is minimized at $a = E[\theta \mid \mathbf{x}]$. ∎

The same argument works for a general **loss function** $L(\theta, \hat\eta)$: the **Bayes risk** $r(\pi, \hat\eta) = E[L(\theta, \hat\eta(\mathbf{X}))]$ is minimized by the estimator that minimizes the **posterior expected loss** $E[L(\theta, \hat\eta) \mid \mathbf{X} = \mathbf{x}]$ for each $\mathbf{x}$. Minimax estimators are defined with the **risk function** $R_{\hat\eta}(\theta) = E[L(\theta, \hat\eta) \mid \theta]$ in place of the MSE.

Example 11.10

(a) Binomial–beta with squared error: $\hat\theta^\pi = \frac{t+\alpha_1}{n+\alpha_1+\alpha_2}$.
(b) Normal model, conjugate prior, loss $L = (\hat\mu-\mu)^2/\sigma^2$: the Bayes estimator minimizes $E[(\hat\mu-\mu)^2\sigma^{-2} \mid \mathbf{x}]$, so $\hat\mu^\pi = E(\mu\sigma^{-2} \mid \mathbf{x})/E(\sigma^{-2} \mid \mathbf{x})$. Since $E(\mu \mid \sigma^2, \mathbf{x}) = \mu_n(\mathbf{x})$ does not depend on σ^2, $E(\mu\sigma^{-2} \mid \mathbf{x}) = \mu_n(\mathbf{x})E(\sigma^{-2} \mid \mathbf{x})$, and $\hat\mu^\pi = \mu_n(\mathbf{x}) = (n\bar x + \kappa_0\mu_0)/(n+\kappa_0)$.
(c) Same model, loss $L = (\hat\sigma^2/\sigma^2 - 1)^2$: the Bayes estimator is $E(\sigma^{-2} \mid \mathbf{x})/E(\sigma^{-4} \mid \mathbf{x})$. With $Y = \nu_n\sigma_n^2(\mathbf{x})/\sigma^2 \sim \chi^2(\nu_n)$, $EY = \nu_n$ and $EY^2 = 2\nu_n + \nu_n^2$, so

$$\hat\sigma^{2,\pi} = \nu_n\sigma_n^2(\mathbf{x})\frac{\nu_n}{2\nu_n+\nu_n^2} = \frac{\nu_n}{\nu_n+2}\sigma_n^2(\mathbf{x}),$$

the posterior mode.

Theorem 11.4 (Bayes estimators with constant risk are minimax)

If an estimator $\hat\eta^*$ has (a) constant risk, $R_{\hat\eta^*}(\theta) = R_{\hat\eta^*}(\theta_0)$ for all θ, and (b) is the Bayes estimator for some prior probability distribution π, then it is minimax.

Proof. By (a), $\max_\theta R_{\hat\eta^*}(\theta) = \int R_{\hat\eta^*}(\theta)\pi(\theta)\,d\theta = r(\pi, \hat\eta^*)$. By (b), for any $\hat\eta$,

$$r(\pi,\hat\eta^*) \le r(\pi,\hat\eta) = \int R_{\hat\eta}(\theta)\pi(\theta)\,d\theta \le \max_\theta R_{\hat\eta}(\theta).$$

Hence $\max_\theta R_{\hat\eta^*} \le \max_\theta R_{\hat\eta}$ for every $\hat\eta$. ■

Example 11.11 (Minimax estimation of a binomial proportion)

For Bernoulli(θ) data and squared error, consider estimators $a\hat\theta + b$ with $\hat\theta = t/n$. Their risk is

$$E[(a\hat\theta+b-\theta)^2 \mid \theta] = \frac{a^2\theta(1-\theta)}{n} + \{b+(a-1)\theta\}^2,$$

a quadratic in θ whose coefficients of θ^2 and θ vanish when $(1-a)^2 = a^2/n$ and $a^2/n - 2b(1-a) = 0$, i.e.

$$a = \frac{\sqrt n}{\sqrt n+1}, \qquad b = \frac{1}{2(\sqrt n+1)}.$$

The resulting estimator $\hat\theta^* = \frac{t+\sqrt n/2}{n+\sqrt n}$ has constant risk $\frac{1}{4(\sqrt n+1)^2}$, and by Example 11.10(a) it is the Bayes estimator for the prior Beta($\sqrt n/2, \sqrt n/2$). By Theorem 11.4 it is minimax. Its maximum risk is smaller than that of $\hat\theta$, which is $1/(4n)$, but $\hat\theta$ has smaller risk except for θ near $1/2$.

When the minimax estimator is not Bayes for any proper prior, it may still be a limit of Bayes estimators.

Theorem 11.5 (Least favourable sequence of priors)

If for some estimator $\hat\eta^*$ there are priors π_k with Bayes estimators $\hat\eta^{\pi_k}$ such that

$$\lim_{k\to\infty} r(\pi_k, \hat\eta^{\pi_k}) \ge \max_\theta R_{\hat\eta^*}(\theta),$$

then $\hat\eta^*$ is minimax.

Proof. For any $\hat\eta$ and every k, $r(\pi_k,\hat\eta^{\pi_k}) \le r(\pi_k,\hat\eta) = \int R_{\hat\eta}\pi_k \le \max_\theta R_{\hat\eta}(\theta)$. Letting $k\to\infty$, $\max_\theta R_{\hat\eta} \ge \lim_k r(\pi_k,\hat\eta^{\pi_k}) \ge \max_\theta R_{\hat\eta^*}$. ■

Example 11.12 (The sample mean is minimax)

For $N(\mu, \sigma^2)$ data and loss $(\hat{\mu}-\mu)^2/\sigma^2$, the Bayes estimator under the conjugate prior is $\mu_n(\mathbf{x})$ (Example 11.10(b)). Given σ^2 and $\mathbf{x}$, $\mu \sim N(\mu_n(\mathbf{x}), \sigma^2/\kappa_n)$, so $E[(\mu - \mu_n(\mathbf{x}))^2/\sigma^2 \mid \sigma^2, \mathbf{x}] = 1/\kappa_n$ and the Bayes risk is $1/(n+\kappa_0)$. Take $\kappa_0 = 1/k$; then $r(\pi_k, \hat{\mu}^{\pi_k}) = 1/(n+1/k) \to 1/n$. The sample mean has constant risk $E[(\bar{X} - \mu)^2/\sigma^2] = 1/n$. By Theorem 11.5, $\bar{X}$ is minimax for this loss.